

10.1 DC: Practical Circuits & Kirchhoff's Laws

Question Paper

Course	CIEA Level Physics
Section	10. D.C. Circuits
Topic	10.1 DC: Practical Circuits & Kirchhoff's Laws
Difficulty	Easy

Time allowed: 40

Score: /32

Percentage: /100

Question 1a

Three resistors of resistances R_1 , R_2 and R_3 are connected as shown in Fig. 1.1.

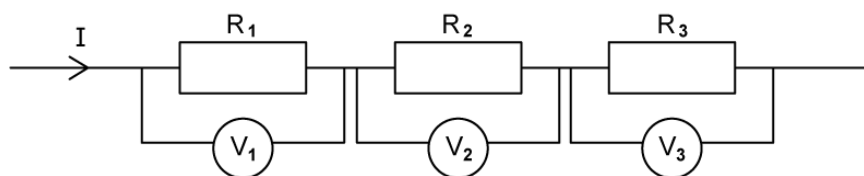


Fig 1.1

The potential difference across the resistors are V_1 , V_2 and V_3 . The current in the combination of resistors is I .

Show that the total resistance R of the combination is given by the equation.

$$R = R_1 + R_2 + R_3$$

[3 marks]

Question 1b

Three resistors are placed in a circuit with a cell of 10 V and negligible internal resistance and two switches S_1 and S_2 . This is shown in Fig 1.1.

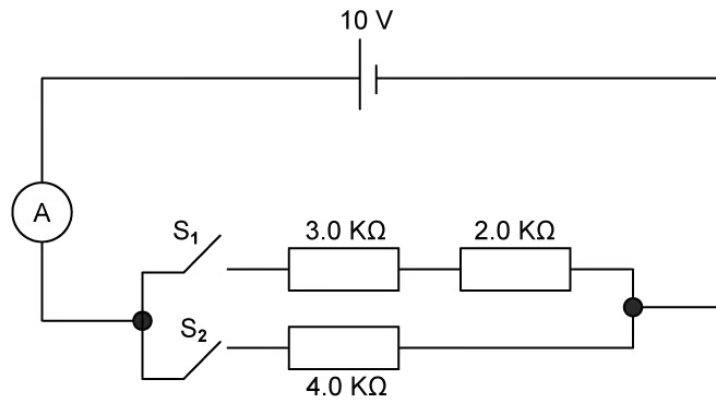


Fig 1.1

Complete table 1.1.

Table 1.1

Position of switches		Total resistance / $k\Omega$
S_1	S_2	
open	closed	
closed	open	
closed	closed	

[3 marks]

Question 1c

The cell has an e.m.f of 10 V and negligible internal resistance.

Calculate the current from the cell when

(i)
switch S_1 is closed,

[1]

(ii)
both S_1 and S_2 are closed.

[1]

[2 marks]

Question 1d

Using your answers in part (c), determine the current from the cell if just switch S_2 is closed.

[3 marks]

Question 2a

Explain why the electromotive force (e.m.f) of a cell with internal resistance may be more than the terminal potential difference (p.d.) of the cell.

[2 marks]

Question 2b

Compare the definitions of e. m .f and p. d.

[3 marks]

Question 2c

For a cell, explain the term *internal resistance*.

[1 mark]

Question 2d

A battery of e.m.f 1508 V and internal resistance r is connected to a resistor A, as shown in Fig 1.1.

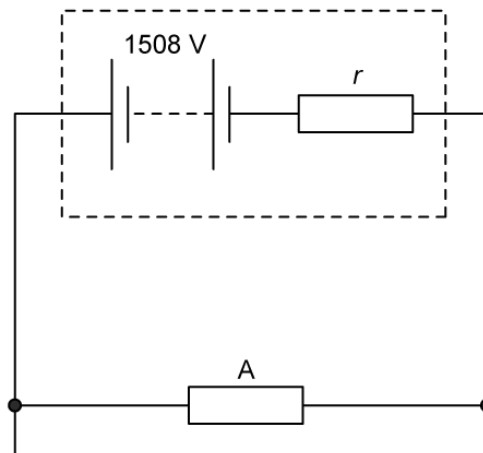


Fig 1.1

The current in resistor A is 400 mA and the power dissipated by it is 600 W.

Calculate r of the battery.

[3 marks]

Question 3a

(i)

State Kirchoff's first law.

[1]

(ii)

State the quantity that is conserved by this law.

[1]

[2 marks]**Question 3b**

(i)

State Kirchoff's second law.

[1]

(ii)

State the quantity that is conserved by this law.

[1]

[2 marks]

Question 3c

Fig 1.1 shows a list of circuit components.

Earth	
Variable resistor	
Heater	
LED	
Thermistor	
Microphone	

Fig 1.1

In the right hand column, sketch the circuit symbol for the component in the left hand column of Fig 1.1.

[6 marks]

Question 3d

Two cells of e.m.f. E_1 and E_2 are negligible internal resistance are connected to resistors R in a circuit shown in Fig 1.2.

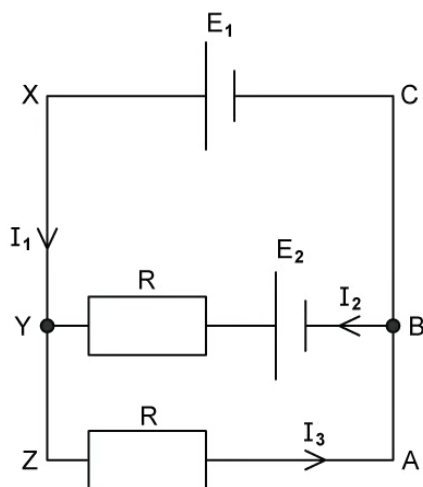


Fig 1.2

Use Kirchhoff's laws to state the relation between

(i)

I_1 , I_2 and I_3

[1]

(ii)

E_2 , R , I_2 and I_3 in loop YZABY.

[1]

[2 marks]

Model Answers: Easy

1a

The total resistance R is given by the equation $R = R_1 + R_2 + R_3$ because:

- The total potential difference in the circuit, V is equal to the sum of the potential differences across each resistor:
 - $V = V_1 + V_2 + V_3$ [1 mark]

Consider Ohm's Law and substitute into the expression for current, V :

- $IR = IR_1 + IR_2 + IR_3$ [1 mark]
- $R = R_1 + R_2 + R_3$ [1 mark]

[Total: 3 marks]

1b

A complete Table 1.1 should look like:

Position of switches		
S_1	S_2	Total resistance / $k\Omega$
open	closed	4.0
closed	open	5.0
closed	closed	2.2

One mark for each correct row

[Total: 3 marks]

If you're struggling with the total resistances when both switches are closed, split the circuit into parallel and series sections.

1. First, combine the series resistors. This is $3.0 + 2.0 = 5.0 k\Omega$. This acts like 1

resistor with a resistance of $5.0 \text{ k}\Omega$.

2. This is now in parallel with the $4.0 \text{ k}\Omega$ resistor which is calculated in parallel. i.e.

$$\frac{1}{R_T} = \frac{1}{5.0} + \frac{1}{4.0} = 0.45. R_T = \frac{1}{0.45} = 2.2 \text{ k}\Omega$$

1c

The current when

(i) switch S_1 is closed is:

List the known quantities:

- Total resistance, $R = 5.0 \text{ k}\Omega = 5.0 \times 10^3 \Omega$ (from part b)

Calculate the current:

- $V = IR \Rightarrow I = \frac{V}{R}$
- $I = \frac{10}{5.0 \times 10^3} = 2 \times 10^{-3} \text{ A (2 mA)}$ [1 mark]

(i) switch S_1 and S_2 are closed is:

List the known quantities:

- Total resistance, $R = 2.2 \text{ k}\Omega = 2.2 \times 10^3 \Omega$ (from part b)

Calculate the current:

- $I = \frac{10}{2.2 \times 10^3} = 4.545 \times 10^{-3} = 4.5 \times 10^{-3} \text{ A (4.5 mA)}$ [1 mark]

[Total: 2 marks]

Due to Kirchhoff's second law, the total potential difference across the branch with S_1 and S_2 is the same. Therefore, V in the equation for both is 10 V .

1d

The current from the cell if switch S_2 is closed is:

List the known quantities:

- The current when S_1 is closed, $I_1 = 2.0 \times 10^{-3} \text{ A}$ (from part c)
- The current when S_1 and S_2 is closed, $I = 4.5 \times 10^{-3} \text{ A}$ (from part c)

Calculate the current when S_2 is closed, I_2 :

- Kirchhoff's first law: $I = I_1 + I_2$ [1 mark]
- $I_2 = I - I_1 = (4.5 \times 10^{-3}) - (2.0 \times 10^{-3})$ [1 mark]
- $I_2 = 2.5 \times 10^{-3} \text{ A}$ (2.5 mA) [1 mark]

[Total: 3 marks]

Although you can get this same answer by doing

$V = \frac{I}{R} = \frac{10}{4.0 \times 10^3} = 2.5 \times 10^{-3} \text{ V}$, the question specifically states by using your answers to part (c) – this implies Kirchhoff's first law. You can, of course, check your answer with Ohm's law, but you cannot gain any marks for reaching an answer this way.

2a

The e.m.f of a cell with internal resistance may be more than the terminal p.d. of the cell because:

- There are lost volts within the cell **OR** energy used / spent on internal resistance; [1 mark]
- When the cell supplies a current; [1 mark]

[Total: 2 marks]

2b

Comparing the definitions of e.m.f and p.d.:

- E.m.f is (the transfer of) chemical energy into electrical energy; [1 mark]
- Whilst p.d. is (the transfer of) electrical energy into thermal energy; [1 mark]
- Per unit charge; [1 mark]

[Total: 3 marks]

2c

Internal resistance is:

- The resistance of a cell causing loss of voltage / energy loss; [1 mark]

[Total: 1 mark]

2d

r of the battery is:

List the known quantities:

- Electromotive force, $E = 1508 \text{ V}$
- Current through resistor A, $I = 400 \text{ mA} = 0.4 \text{ A}$
- Power dissipated, $P = 600 \text{ W}$

Calculate the potential difference V through resistor A:

- $P = IV \Rightarrow V = \frac{P}{I}$
- $V = \frac{600}{0.4} = 1500 \text{ V}$ [1 mark]

Calculate the internal resistance, r .

- $E = V + Ir \Rightarrow r = \frac{E - V}{I}$
- $r = \frac{1508 - 1500}{0.4}$ [1 mark]
- $r = 20 \Omega$ [1 mark]

[Total: 3 marks]

3a

(i) Kirchhoff's first law states that:

- The sum of currents into a junctions equals the sum of the currents out of a junction; [1 mark]

(ii) The quantity conserved by this law is:

- Charge; [1 mark]

[Total: 2 marks]

It is common to mistakenly say that 'current' is conserved in this law. Although that's true, current is just the flow of **charge**, which is the fundamental conservation linked with Kirchhoff's first law. Therefore, for part (ii), 'current' will **not** get the mark.

3b

(i) Kirchhoff's second law states that:

- The sum of e.m.f's equals the sum of the p.d's around a loop / circuit; [1 mark]

(ii) The quantity conserved by this law is:

- Energy; [1 mark]

[Total: 2 marks]

It is common to mistakenly say that 'p.d' is conserved in this law. However, p.d. is the measure of energy per unit charge transferred between two points in a circuit. Therefore, it is technically **energy** that is conserved.

3c

The correct circuit symbols are:

junction; [1 mark]

(ii) The quantity conserved by this law is:

- Charge; [1 mark]

[Total: 2 marks]

It is common to mistakenly say that 'current' is conserved in this law. Although that's true, current is just the flow of **charge**, which is the fundamental conservation linked with Kirchhoff's first law. Therefore, for part (ii), 'current' will **not** get the mark.

3b

(i) Kirchhoff's second law states that:

- The sum of e.m.f's equals the sum of the p.d's around a loop / circuit; [1 mark]

(ii) The quantity conserved by this law is:

- Energy; [1 mark]

[Total: 2 marks]

It is common to mistakenly say that 'p.d' is conserved in this law. However, p.d. is the measure of energy per unit charge transferred between two points in a circuit. Therefore, it is technically **energy** that is conserved.

3c

The correct circuit symbols are:

Earth	
Variable resistor	

Heater	
LED	
Thermistor	
Microphone	

- *One mark for each correct circuit symbol*

[Total: 6 marks]

There are some niche circuit symbols required in this question, however, they are all required according to the CIE IAL syllabus. The end of the syllabus has a list of all the circuit symbols that you should recognise in your exam questions, make sure to familiarise yourself with **all** of them.

3d

The relation between:

(i) I_1 , I_2 and I_3 is:

- $I_1 + I_2 = I_3$ [1 mark]

(ii) E_2 , R , I_2 and I_3 in loop YZABY is:

- $E_2 = I_2 R + I_3 R$ [1 mark]

[Total: 2 marks]

Remember from Ohm's law that $V = IR$.

Part (i) is using Kirchhoff's first law, whilst part (ii) is using Kirchhoff's second law.

